



Onset of shear thinning and thickening in frictionless granular system

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ABSTRACT

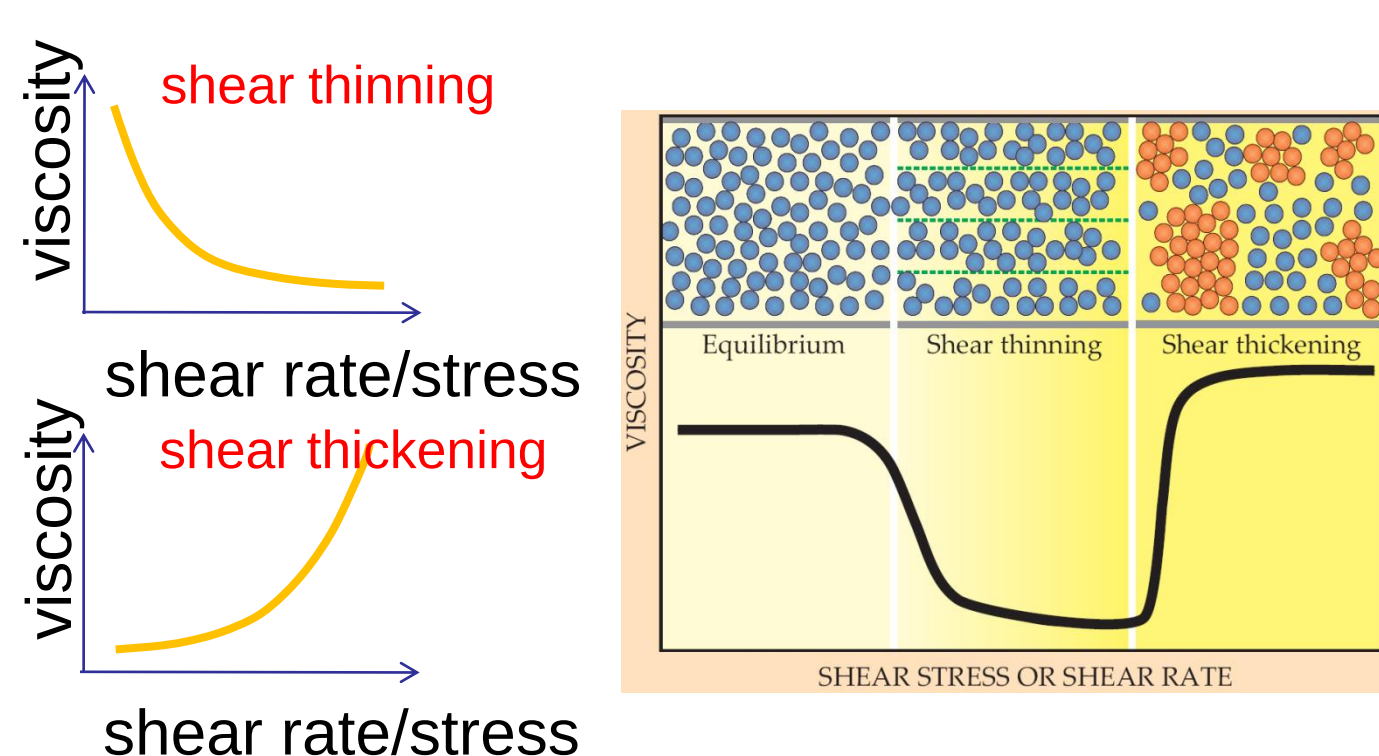
We perform non-equilibrium molecular dynamics simulations to analyze the rheology of two-dimensional frictionless granular materials. We find that the steady-flow rheological data show four regimes as a function of increasing shear rate, which can be divided into Newtonian, shear-thinning I, shear thickening, and shear-thinning II, respectively. The magnitude of shear thickening is enhanced as the system size increase and becomes a -function located in the infinite system size limit. Before the transition from shear thinning to shear thickening, a strong stress overshoot is observed upon start-up of flow, and its magnitude greatly depended on the shear rate. From the structural point of view, the maximum of the pair distribution function decreases as shear rate increases in shear thinning region, and is sitting at a minimum in the shear thickening region.

INTRODUCTION

- Newtonian fluids (like water):



- For most Non-Newtonian fluids, the viscosity decreases with increasing shear rate, that is, they shear thin. Few are reported to do the opposite, that is, shear thicken, despite the longstanding expectation that shear thickening is a generic type of Non-Newtonian behavior.

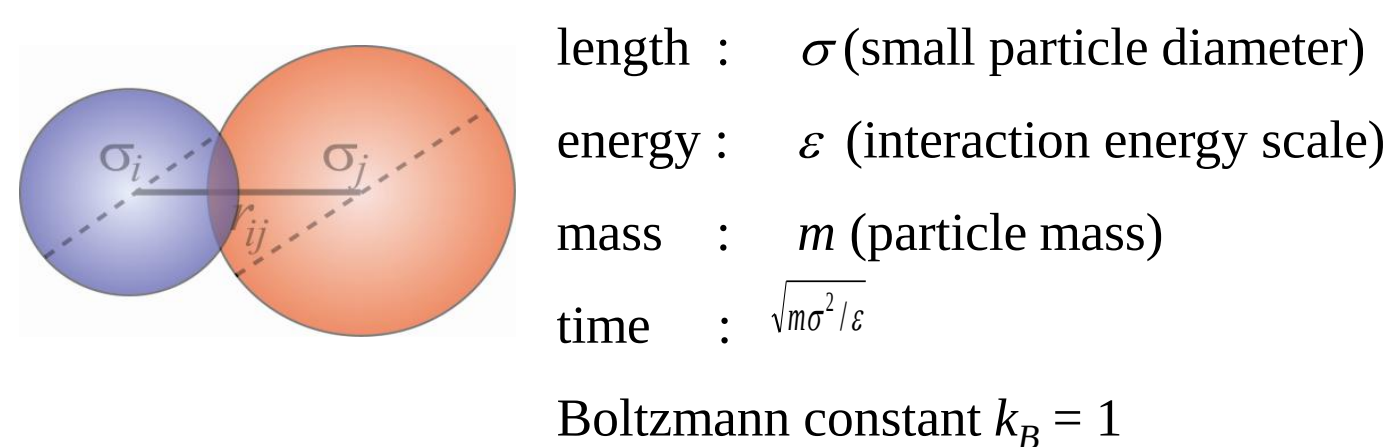


- Shear thickening is presumed to be due to general mechanisms such as hydrodynamics or dilation.

MODEL OF PACKING OF SPHERES

- $N/2$ big and $N/2$ small frictionless spheres with mass m Cubic box with periodic boundary conditions $\sigma_L / \sigma_S = 1.4 \rightarrow$ avoid crystallization

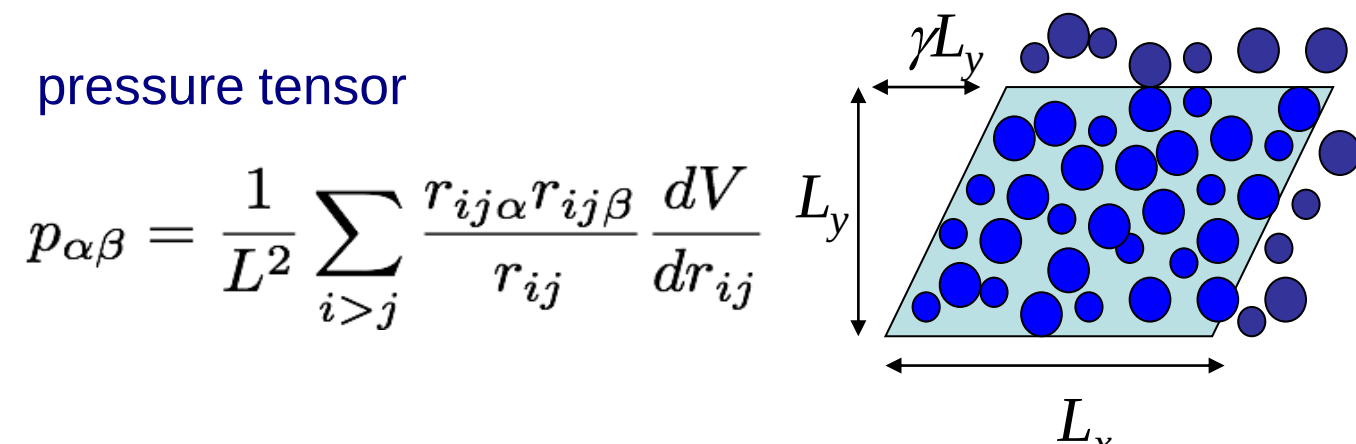
- Purely repulsive interactions



$$V(r_{ij}) = \begin{cases} \varepsilon(1 - r_{ij}/\sigma_{ij})^\alpha / \alpha, & r_{ij} < \sigma_{ij} \\ 0, & r_{ij} > \sigma_{ij} \end{cases}$$

Harmonic: $\alpha=2$; Hertzian: $\alpha=5/2$

- Lees-Edwards boundary Conditions for shear flow

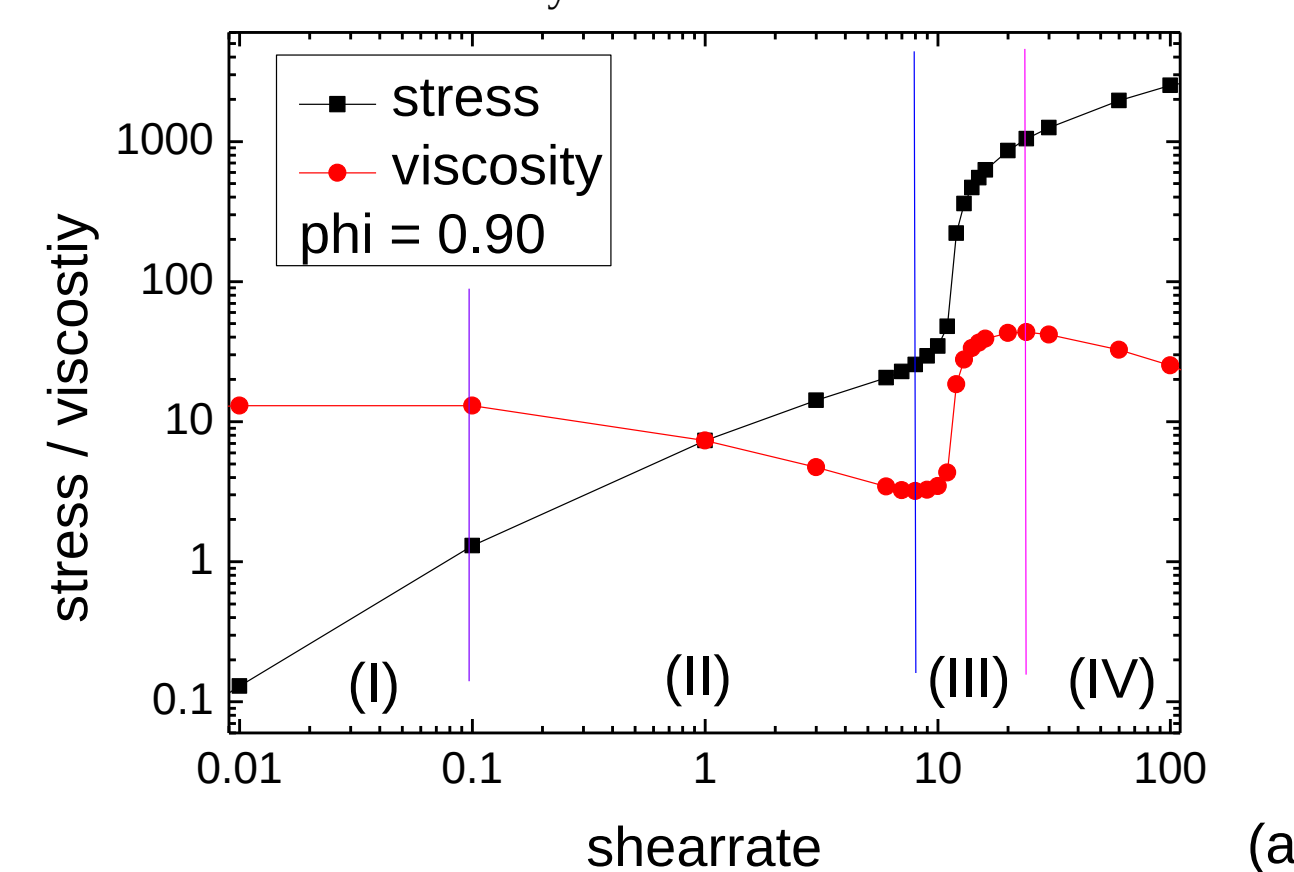


FLOW CURVES AT DIFFERENT SHEARRATE

- Shear stress and shear viscosity:

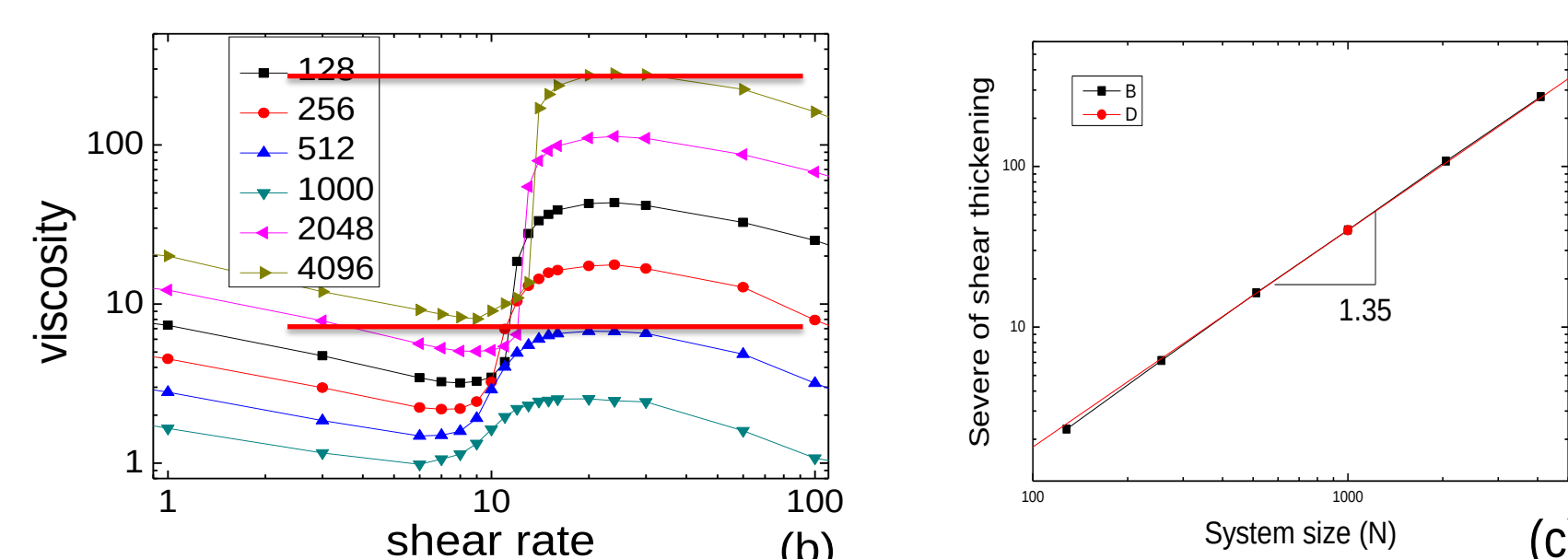
$$\text{force: } \vec{F}_i = \sum_j [\vec{F}_{ij}^r(r_{ij}) - b(\vec{v}_i - \vec{v}_j) \cdot \hat{r}_{ij} \hat{r}_{ij}]$$

$$\text{stress: } \Sigma = - \frac{\sum_{i \neq j}^{N_0} x_{ij} F_{yij}}{2L_x L_y} \quad \text{viscosity: } \eta = \frac{\Sigma}{\dot{\gamma}}$$



(I)Newtonian (II)shear thinning I (III)shear thickening (IV)shear thinning II

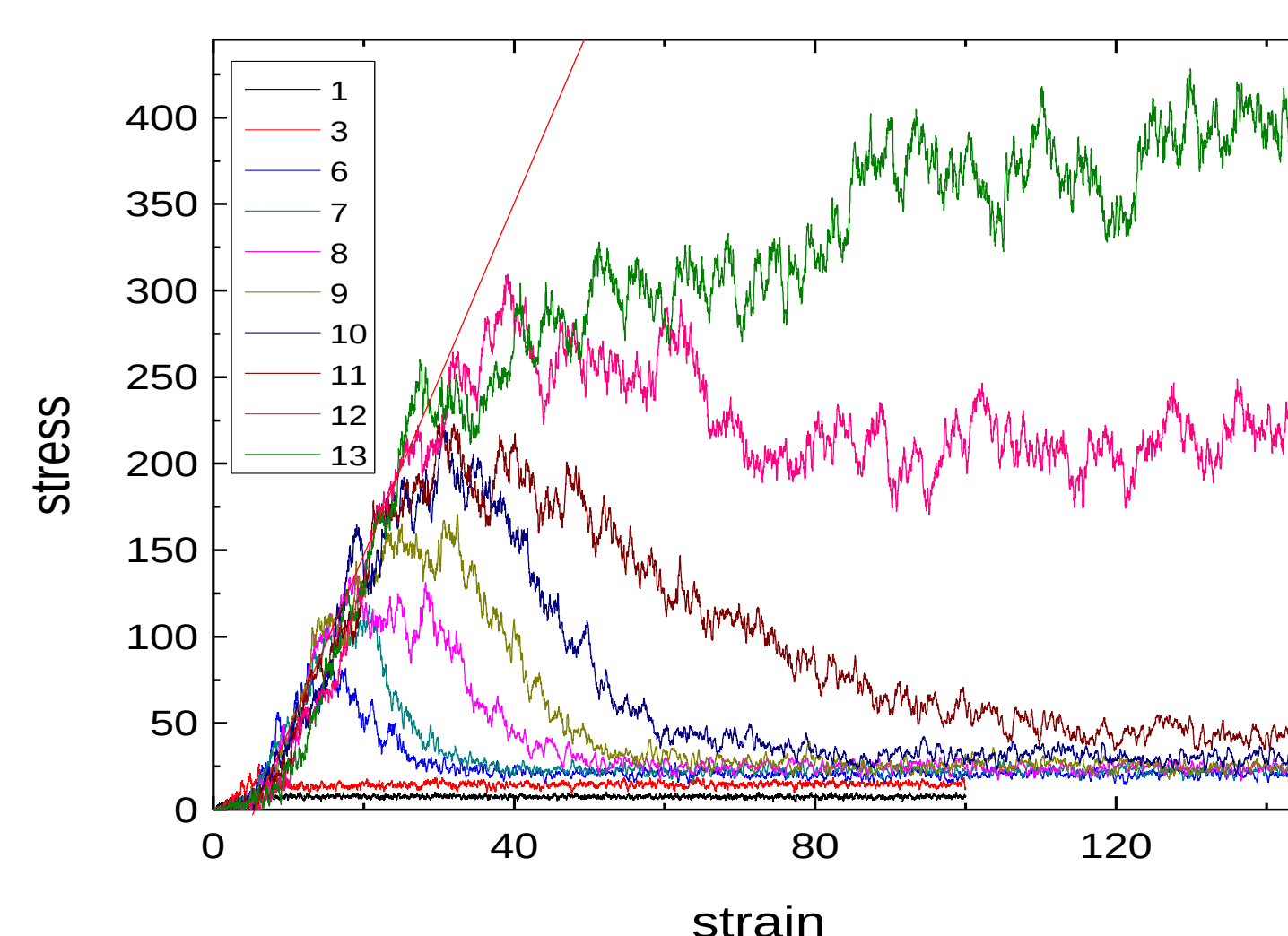
- The influence of system size



- As N increases, the magnitude of shear thickening increase.
- Log-log plot of the data of magnitude of shear thickening vs. system size. The dashed line has slope 1.35.

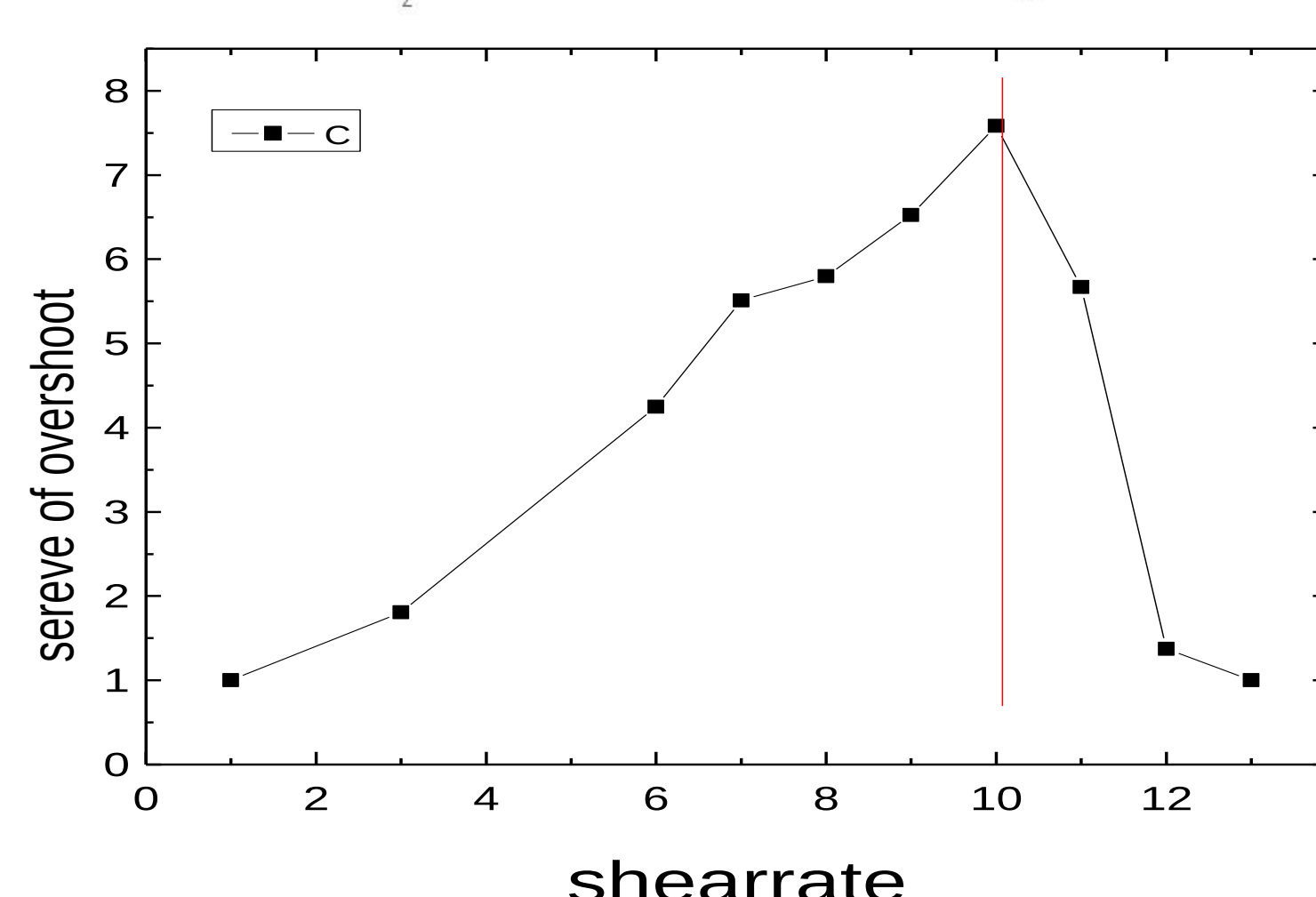
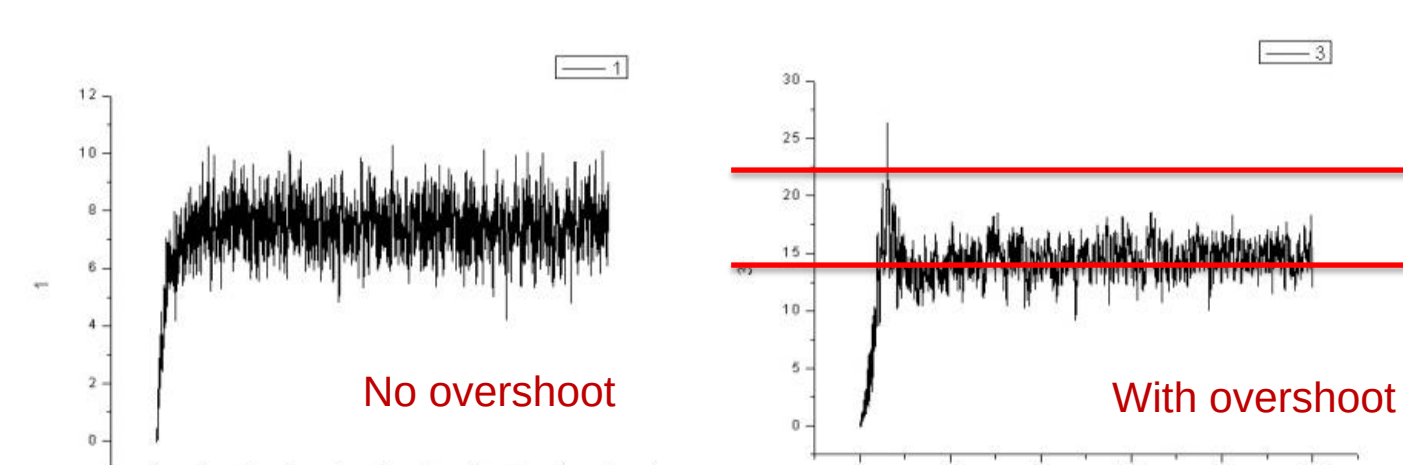
STRESS OVERSHOOT ON STARTUP

- start-up of flow curve at different applied shear rate



Stress vs. strain, during start-up at for different applied shear rate, for (shear rate increasing with increasing size of overshoot).

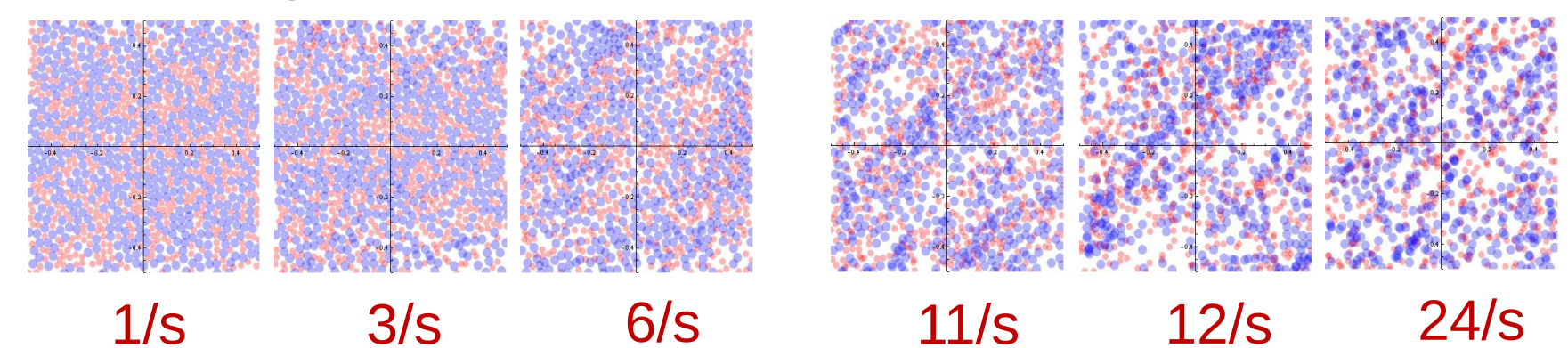
- Stress normalized by the peak stress



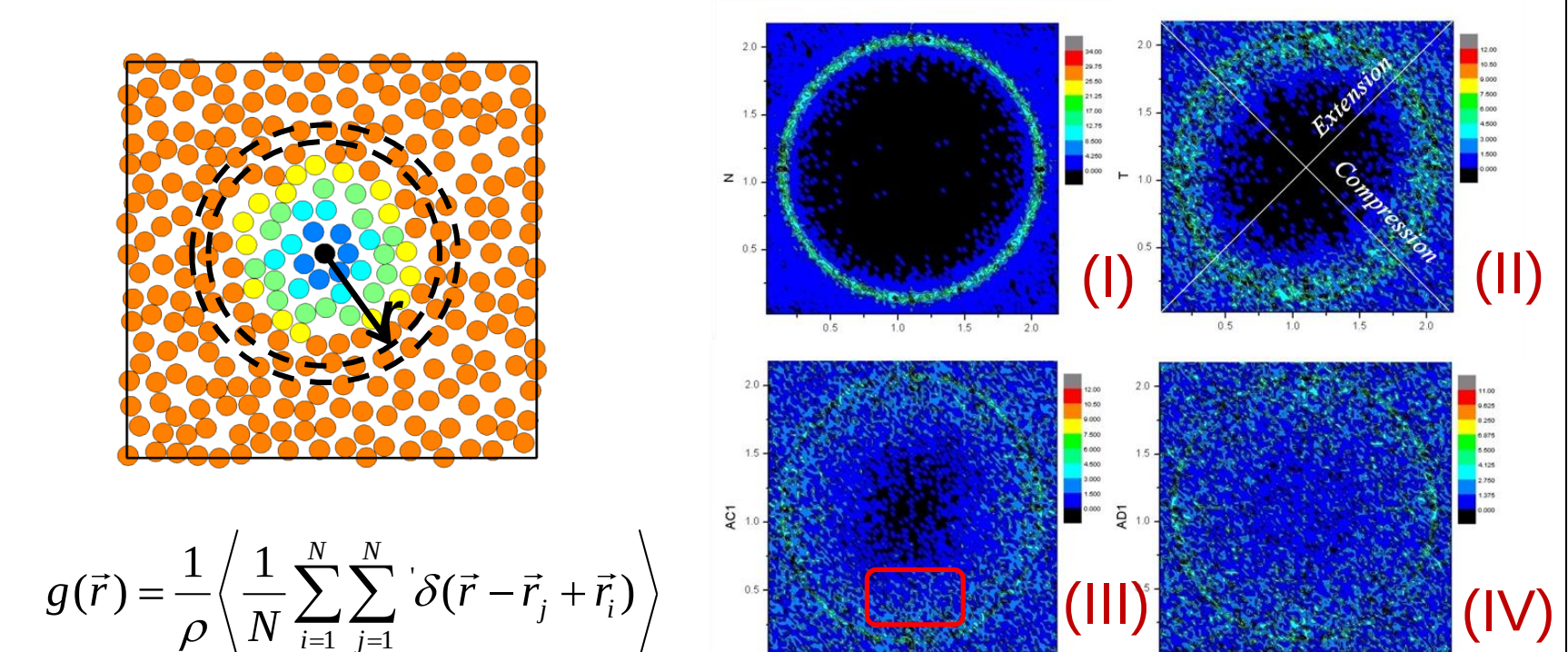
- Stress normalized by the peak stress, Σ_k / Σ , at different shear rate.

STRUCTURE OF PACKING OF SPHERES UNDER SHEAR

- configuration under various shear rate

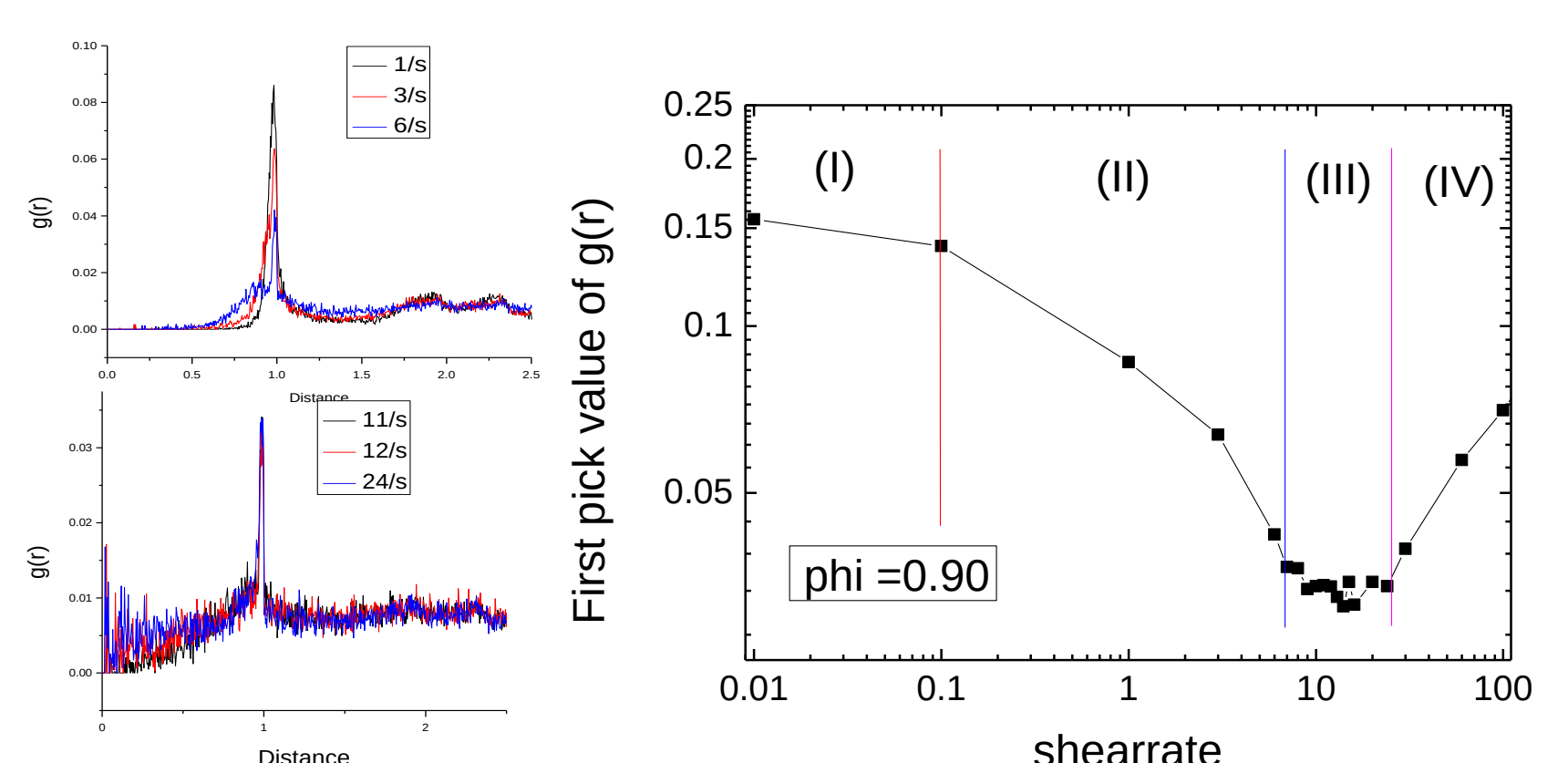


- 2D radial distribution function



- 1D radial distribution function $g(r)$

$$g(r) = \frac{L^d}{N^2} \left\langle \sum_i \sum_j \delta(\vec{r} - \vec{r}_{ij}) \right\rangle$$



(I)Newtonian (II)shear thinning I (III)shear thickening (IV)shear thinning II

- The first pick value of $g(r)$ is decrease as the shear rate increasing in the shear thinning area, and is at the low value in the shear thickening area.
- Cluster if not mean the shear thickening, is also appear in the shear thinning area.

CONCLUSIONS

- Flow curve:** Steady-flow rheological data show four regimes as a function of increasing shear rate, which can be divided into Newtonian, shear-thinning I, shear thickening, and shear-thinning II, respectively.
- System size:** The magnitude of shear thickening is enhanced as the system size increase and becomes a -function located in the infinite system size limit.
- Stress overshoot:** Before the transition from shear thinning to shear thickening, a strong stress overshoot is observed upon start-up of flow, and its magnitude greatly depended on the shear rate.
- Structure:** From the structural point of view, the maximum of the pair distribution function decreases as shear rate increases in shear thinning region, and is sitting at a minimum in the shear thickening region.

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